

7.2:

$$\text{base } C = (1, x, x^2)$$

linearly
independent

$$\phi(1) = 0$$

$$\phi(x) = 2$$

$$\phi(x^2) = 4x + 2x^2$$

matrix ϕ w.r.t. C :

$$A = \begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 2 \end{pmatrix}$$

diagonalizable matrix

$\lambda_1 = 0$ algebraic multiplicity 2
geom. multiplicity 1

$\lambda_2 = 2$ multiplicity 1

$$\begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

Jordan form:

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$\lambda = 0$ kernel non-trivial

$$N_2 \Rightarrow N_1 \Rightarrow 0$$

$$\phi(f(x)) = 0 \cdot f(x)$$

$$f = a + bx + cx^2$$

$$f' = b + 2cx$$

$$f'' = 2c$$

$$\phi(f) = 2b + 4cx + 2cx^2 = 0$$

$$f(x) = a$$

$$N_1 = 1$$

N_2 :

$$\phi(N_2) = N_1 = 1$$

$$N_2 = a + bx$$

$$\phi(N_2) = 2b = 1$$

$$b = \frac{1}{2} \Rightarrow N_2 = \frac{x}{2}$$

$$\lambda = 2: \quad \phi(f) = 2f \quad f = a + bx + cx^2$$

$$\phi(f) = 2b + 4cx + 2cx^2 = 2a + 2bx + 2cx^2$$

$$2a = 2b \quad b = a$$

$$4c = 2b \quad 2c = b = a \quad c = \frac{a}{2}$$

$$2c = 2c$$

$$f = a + ax + \frac{a}{2}x^2$$

$$a = 2 \quad f = 2 + 2x + x^2$$

$$[\phi]_B^B =] \quad \text{po } B = \left(1, \frac{x}{2}, 2 + 2x + x^2 \right)$$